

# Supplementary Information

## InFo-DRP: Integrating Invariant Learning and Foundation Representations for Unseen Drug Response Prediction

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**Table S1. Hyperparameter settings for InFo-DRP.** All hyperparameters were fixed across experiments except the learning rate, selected per dataset and evaluation setting from a set of candidate values based on validation performance; early stopping was based on validation RMSE. Learning rates larger than this range gave less stable optimization and lower validation performance in initial experiments, and the search was therefore restricted to the range listed here.

| Category           | Hyperparameter                                      | Value                      |
|--------------------|-----------------------------------------------------|----------------------------|
| Architecture       | Latent Dimension                                    | 256                        |
|                    | Fusion Hidden Dimension                             | 1024                       |
|                    | Transformer Layers                                  | 3                          |
|                    | Attention Heads                                     | 8                          |
|                    | Dropout Rate                                        | 0.1                        |
| Invariant Learning | Codebook Size                                       | 1000                       |
|                    | Loss Weight $\lambda_{\text{inv}}$                  | $1 \times 10^{-4}$         |
|                    | Loss Weight $\lambda_{\text{reg}}$                  | 0.1                        |
|                    | Loss Weight $\lambda_{\text{cmt}}$                  | 0.05                       |
|                    | Mean Gate Target ( $\gamma_{\text{drug}}$ )         | 0.9                        |
|                    | Mean Gate Target ( $\gamma_{\text{cell}}$ )         | 0.7                        |
| Optimization       | Batch Size                                          | 32                         |
|                    | Optimizer                                           | AdamW                      |
|                    | Weight Decay                                        | $1 \times 10^{-4}$         |
|                    | Max Epochs (Early Stop Patience)                    | 300 (10)                   |
|                    | Learning Rate Search Range                          | $[1, 3, 5] \times 10^{-5}$ |
|                    | Learning Rate (GDSC <sub>v1</sub> , random-split)   | $5 \times 10^{-5}$         |
|                    | Learning Rate (GDSC <sub>v1</sub> , cell-blind)     | $5 \times 10^{-5}$         |
|                    | Learning Rate (GDSC <sub>v1</sub> , drug-blind)     | $5 \times 10^{-5}$         |
|                    | Learning Rate (GDSC <sub>v2</sub> , cell-blind)     | $3 \times 10^{-5}$         |
|                    | Learning Rate (GDSC <sub>v2</sub> , drug-blind)     | $5 \times 10^{-5}$         |
|                    | Learning Rate (GDSC <sub>v2</sub> , scaffold-blind) | $1 \times 10^{-5}$         |

**Table S2. Hyperparameter settings for machine learning baselines (GDSC<sub>v2</sub>).** Values represent the optimal settings for each split, selected from a set of candidate hyperparameter configurations based on validation performance. The candidate configurations evaluated for each model were as follows. Random Forest: three configurations with  $(n\_estimators, max\_depth, min\_samples\_leaf) \in \{(500, None, 1), (800, 40, 2), (1000, 30, 1)\}$ . Elastic Net: three configurations with  $(alpha, l1\_ratio) \in \{(10^{-3}, 0.1), (10^{-2}, 0.5), (10^{-1}, 0.9)\}$ . XGBoost: two configurations with  $(n\_estimators, max\_depth, learning\_rate) \in \{(600, 6, 0.1), (800, 8, 0.05)\}$ . LightGBM: two configurations with  $(num\_leaves, learning\_rate) \in \{(63, 0.1), (127, 0.05)\}$  and  $n\_estimators=1000$ . For tree-based boosting models, subsampling-related hyperparameters were fixed ( $subsample=0.8$ ,  $colsample\_bytree=0.8$  for XGBoost;  $feature\_fraction=0.8$ ,  $bagging\_fraction=0.8$  for LightGBM) to control overfitting. Additional settings included  $max\_iter=5000$  for Elastic Net,  $reg\_lambda=1.0$  and  $tree\_method='hist'$  for XGBoost, and  $bagging\_freq=1$  and  $objective='regression'$  for LightGBM.

| Model         | Hyperparameter   | Selected Value |            |                |
|---------------|------------------|----------------|------------|----------------|
|               |                  | cell-blind     | drug-blind | scaffold-blind |
| Random Forest | n_estimators     | 500            | 500        | 1000           |
|               | max_depth        | None           | None       | 30             |
|               | min_samples_leaf | 1              | 1          | 1              |
| Elastic Net   | alpha            | 0.01           | 0.001      | 0.001          |
|               | l1_ratio         | 0.5            | 0.1        | 0.1            |
| XGBoost       | n_estimators     | 800            | 800        | 600            |
|               | learning_rate    | 0.05           | 0.05       | 0.1            |
|               | max_depth        | 8              | 8          | 6              |
| LightGBM      | n_estimators     | 1000           | 1000       | 1000           |
|               | learning_rate    | 0.05           | 0.1        | 0.1            |
|               | num_leaves       | 127            | 63         | 63             |

**Table S3. Extended ablation of molecular embedding branches (GDSC<sub>v1</sub>).** Results for random-split and cell-blind settings.

| Variant         | random-split   |                   | cell-blind     |                   |
|-----------------|----------------|-------------------|----------------|-------------------|
|                 | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ |
| InFo-DRP        | 0.934 (0.0007) | 0.973 (0.0062)    | 0.877 (0.0022) | 1.320 (0.0132)    |
| w/o MolBERT     | 0.934 (0.0010) | 0.972 (0.0066)    | 0.876 (0.0013) | 1.323 (0.0047)    |
| w/o Chemformer  | 0.935 (0.0013) | 0.965 (0.0079)    | 0.877 (0.0002) | 1.320 (0.0050)    |
| w/o ECFP4       | 0.934 (0.0015) | 0.969 (0.0101)    | 0.876 (0.0037) | 1.324 (0.0189)    |
| w/o Descriptors | 0.933 (0.0013) | 0.976 (0.0094)    | 0.876 (0.0022) | 1.319 (0.0114)    |

**Table S4. Extended ablation of architectural components (GDSC<sub>v1</sub>).** Results for random-split and cell-blind settings.

| Variant                 | random-split   |                   | cell-blind     |                   |
|-------------------------|----------------|-------------------|----------------|-------------------|
|                         | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ |
| InFo-DRP                | 0.934 (0.0007) | 0.973 (0.0062)    | 0.877 (0.0022) | 1.320 (0.0132)    |
| w/o Cell Invariant      | 0.933 (0.0010) | 0.977 (0.0066)    | 0.875 (0.0022) | 1.327 (0.0126)    |
| w/o Drug Invariant      | 0.934 (0.0009) | 0.973 (0.0064)    | 0.875 (0.0025) | 1.324 (0.0107)    |
| w/o Both Invariants     | 0.933 (0.0005) | 0.978 (0.0034)    | 0.876 (0.0026) | 1.321 (0.0116)    |
| w/o Projector MLP       | 0.933 (0.0009) | 0.984 (0.0056)    | 0.875 (0.0017) | 1.327 (0.0103)    |
| w/o Transformer encoder | 0.935 (0.0011) | 0.958 (0.0077)    | 0.875 (0.0027) | 1.327 (0.0131)    |

**Table S5. Extended single-branch analysis (GDSC<sub>v1</sub>).** Results for random-split, cell-blind, and drug-blind settings.

| Variant                      | random-split   |                   | cell-blind     |                   | drug-blind     |                   |
|------------------------------|----------------|-------------------|----------------|-------------------|----------------|-------------------|
|                              | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ |
| <i>Modality: MolBERT</i>     |                |                   |                |                   |                |                   |
| BASE                         | 0.936 (0.0011) | 0.954 (0.0073)    | 0.877 (0.0018) | 1.315 (0.0110)    | 0.580 (0.0640) | 2.199 (0.1223)    |
| + Invariant                  | 0.937 (0.0006) | 0.948 (0.0041)    | 0.877 (0.0009) | 1.317 (0.0027)    | 0.565 (0.0238) | 2.243 (0.0278)    |
| <i>Modality: Chemformer</i>  |                |                   |                |                   |                |                   |
| BASE                         | 0.934 (0.0009) | 0.970 (0.0066)    | 0.876 (0.0028) | 1.320 (0.0140)    | 0.475 (0.0277) | 2.525 (0.0944)    |
| + Invariant                  | 0.936 (0.0004) | 0.959 (0.0034)    | 0.875 (0.0021) | 1.328 (0.0119)    | 0.496 (0.0321) | 2.421 (0.0583)    |
| <i>Modality: ECFP4</i>       |                |                   |                |                   |                |                   |
| BASE                         | 0.937 (0.0005) | 0.949 (0.0044)    | 0.877 (0.0026) | 1.320 (0.0137)    | 0.545 (0.0377) | 2.245 (0.0534)    |
| + Invariant                  | 0.937 (0.0011) | 0.946 (0.0083)    | 0.877 (0.0009) | 1.315 (0.0084)    | 0.543 (0.0526) | 2.255 (0.0627)    |
| <i>Modality: Descriptors</i> |                |                   |                |                   |                |                   |
| BASE                         | 0.935 (0.0008) | 0.962 (0.0075)    | 0.877 (0.0021) | 1.320 (0.0121)    | 0.540 (0.0226) | 2.283 (0.0367)    |
| + Invariant                  | 0.936 (0.0012) | 0.958 (0.0076)    | 0.878 (0.0028) | 1.314 (0.0131)    | 0.523 (0.0327) | 2.335 (0.0939)    |

**Table S6. Ablation of paired molecular embedding branches (GDSC<sub>v1</sub>).** Results for random-split, cell-blind, and drug-blind settings. Each variant removes the two listed branches and retains the remaining two.

| Variant                      | random-split   |                   | cell-blind     |                   | drug-blind     |                   |
|------------------------------|----------------|-------------------|----------------|-------------------|----------------|-------------------|
|                              | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ | PCC $\uparrow$ | RMSE $\downarrow$ |
| InFo-DRP                     | 0.934 (0.0007) | 0.973 (0.0062)    | 0.877 (0.0022) | 1.320 (0.0132)    | 0.631 (0.0231) | 2.106 (0.0429)    |
| w/o MolBERT + Chemformer     | 0.936 (0.0007) | 0.959 (0.0049)    | 0.875 (0.0020) | 1.324 (0.0113)    | 0.553 (0.0634) | 2.264 (0.1183)    |
| w/o MolBERT + ECFP4          | 0.935 (0.0009) | 0.961 (0.0067)    | 0.876 (0.0017) | 1.323 (0.0082)    | 0.560 (0.0428) | 2.345 (0.1784)    |
| w/o MolBERT + Descriptors    | 0.934 (0.0012) | 0.971 (0.0086)    | 0.876 (0.0009) | 1.318 (0.0045)    | 0.599 (0.0148) | 2.156 (0.0269)    |
| w/o Chemformer + ECFP4       | 0.934 (0.0013) | 0.967 (0.0086)    | 0.876 (0.0013) | 1.321 (0.0048)    | 0.601 (0.0223) | 2.174 (0.0391)    |
| w/o Chemformer + Descriptors | 0.934 (0.0008) | 0.966 (0.0042)    | 0.878 (0.0013) | 1.312 (0.0062)    | 0.605 (0.0211) | 2.149 (0.0393)    |
| w/o ECFP4 + Descriptors      | 0.934 (0.0015) | 0.973 (0.0090)    | 0.876 (0.0030) | 1.321 (0.0163)    | 0.562 (0.0433) | 2.275 (0.1060)    |

**Table S7. PCC across scaffold similarity bins (GDSC<sub>v2</sub>, scaffold-blind setting).** Bin counts (N) correspond to the number of test instances in each scaffold similarity interval. Values are reported as mean  $\pm$  SEM.

| Method         | 0.0–0.2 (N=4681)  | 0.2–0.4 (N=77715) | 0.4–0.6 (N=78332) | 0.6–0.8 (N=16612) | 0.8–1.0 (N=1859)  |
|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Random Forest  | 0.214 $\pm$ 0.248 | 0.362 $\pm$ 0.052 | 0.252 $\pm$ 0.076 | 0.765 $\pm$ 0.042 | 0.511 $\pm$ 0.205 |
| Elastic Net    | 0.492 $\pm$ 0.175 | 0.445 $\pm$ 0.026 | 0.368 $\pm$ 0.047 | 0.654 $\pm$ 0.105 | 0.632 $\pm$ 0.105 |
| XGBoost        | 0.431 $\pm$ 0.285 | 0.386 $\pm$ 0.048 | 0.373 $\pm$ 0.055 | 0.696 $\pm$ 0.089 | 0.658 $\pm$ 0.072 |
| LightGBM       | 0.423 $\pm$ 0.251 | 0.409 $\pm$ 0.049 | 0.382 $\pm$ 0.037 | 0.734 $\pm$ 0.069 | 0.638 $\pm$ 0.094 |
| DeepCDR (Expr) | 0.651 $\pm$ 0.090 | 0.417 $\pm$ 0.068 | 0.396 $\pm$ 0.045 | 0.751 $\pm$ 0.052 | 0.637 $\pm$ 0.067 |
| DeepTTA        | 0.726 $\pm$ 0.081 | 0.354 $\pm$ 0.074 | 0.309 $\pm$ 0.039 | 0.651 $\pm$ 0.102 | 0.654 $\pm$ 0.095 |
| InFo-DRP       | 0.691 $\pm$ 0.056 | 0.534 $\pm$ 0.046 | 0.571 $\pm$ 0.051 | 0.741 $\pm$ 0.067 | 0.595 $\pm$ 0.027 |

**Table S8. RMSE across scaffold similarity bins (GDSC<sub>v2</sub>, scaffold-blind setting).** Bin counts (N) correspond to the number of test instances in each scaffold similarity interval. Values are reported as mean  $\pm$  SEM.

| Method         | 0.0–0.2 (N=4681)  | 0.2–0.4 (N=77715) | 0.4–0.6 (N=78332) | 0.6–0.8 (N=16612) | 0.8–1.0 (N=1859)  |
|----------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Random Forest  | 5.451 $\pm$ 1.166 | 2.649 $\pm$ 0.244 | 3.065 $\pm$ 0.187 | 1.648 $\pm$ 0.264 | 3.997 $\pm$ 1.411 |
| Elastic Net    | 4.851 $\pm$ 1.624 | 2.253 $\pm$ 0.126 | 2.526 $\pm$ 0.255 | 1.999 $\pm$ 0.251 | 3.185 $\pm$ 1.491 |
| XGBoost        | 4.659 $\pm$ 1.606 | 2.323 $\pm$ 0.146 | 2.686 $\pm$ 0.273 | 1.763 $\pm$ 0.212 | 3.642 $\pm$ 2.354 |
| LightGBM       | 4.402 $\pm$ 1.691 | 2.372 $\pm$ 0.174 | 2.707 $\pm$ 0.231 | 1.679 $\pm$ 0.219 | 3.644 $\pm$ 2.202 |
| DeepCDR (Expr) | 4.764 $\pm$ 1.971 | 2.352 $\pm$ 0.184 | 2.751 $\pm$ 0.264 | 1.713 $\pm$ 0.203 | 3.383 $\pm$ 0.181 |
| DeepTTA        | 3.467 $\pm$ 1.486 | 2.287 $\pm$ 0.099 | 2.663 $\pm$ 0.260 | 1.716 $\pm$ 0.139 | 3.132 $\pm$ 0.406 |
| InFo-DRP       | 3.822 $\pm$ 1.880 | 2.131 $\pm$ 0.165 | 2.192 $\pm$ 0.179 | 1.398 $\pm$ 0.073 | 3.140 $\pm$ 0.180 |

**Table S9. PCC across drug similarity bins (GDSC<sub>v2</sub>, drug-blind setting).** Bin counts (N) correspond to the number of test instances in each drug similarity interval. Values are reported as mean  $\pm$  SEM.

| Method         | 0.0–0.2 (N=9544)  | 0.2–0.4 (N=125333) | 0.4–0.6 (N=27443) | 0.6–0.8 (N=15757) | 0.8–1.0 (N=2581)  |
|----------------|-------------------|--------------------|-------------------|-------------------|-------------------|
| Random Forest  | 0.192 $\pm$ 0.187 | 0.268 $\pm$ 0.088  | 0.283 $\pm$ 0.143 | 0.724 $\pm$ 0.043 | 0.794 $\pm$ 0.014 |
| Elastic Net    | 0.480 $\pm$ 0.149 | 0.356 $\pm$ 0.056  | 0.289 $\pm$ 0.121 | 0.659 $\pm$ 0.089 | 0.631 $\pm$ 0.062 |
| XGBoost        | 0.400 $\pm$ 0.174 | 0.404 $\pm$ 0.064  | 0.342 $\pm$ 0.129 | 0.790 $\pm$ 0.041 | 0.791 $\pm$ 0.023 |
| LightGBM       | 0.349 $\pm$ 0.158 | 0.362 $\pm$ 0.074  | 0.325 $\pm$ 0.144 | 0.782 $\pm$ 0.035 | 0.794 $\pm$ 0.026 |
| DeepCDR (Expr) | 0.673 $\pm$ 0.051 | 0.352 $\pm$ 0.060  | 0.352 $\pm$ 0.111 | 0.664 $\pm$ 0.090 | 0.568 $\pm$ 0.083 |
| DeepTTA        | 0.405 $\pm$ 0.123 | 0.336 $\pm$ 0.040  | 0.465 $\pm$ 0.109 | 0.628 $\pm$ 0.068 | 0.652 $\pm$ 0.072 |
| InFo-DRP       | 0.593 $\pm$ 0.069 | 0.521 $\pm$ 0.051  | 0.561 $\pm$ 0.112 | 0.834 $\pm$ 0.027 | 0.718 $\pm$ 0.036 |

**Table S10. RMSE across drug similarity bins (GDSC<sub>v2</sub>, drug-blind setting).** Bin counts (N) correspond to the number of test instances in each drug similarity interval. Values are reported as mean  $\pm$  SEM.

| Method         | 0.0–0.2 (N=9544)  | 0.2–0.4 (N=125333) | 0.4–0.6 (N=27443) | 0.6–0.8 (N=15757) | 0.8–1.0 (N=2581)  |
|----------------|-------------------|--------------------|-------------------|-------------------|-------------------|
| Random Forest  | 3.994 $\pm$ 0.807 | 2.815 $\pm$ 0.268  | 2.862 $\pm$ 0.542 | 2.851 $\pm$ 0.444 | 1.663 $\pm$ 0.417 |
| Elastic Net    | 3.268 $\pm$ 0.954 | 2.429 $\pm$ 0.154  | 2.298 $\pm$ 0.217 | 2.671 $\pm$ 0.196 | 2.197 $\pm$ 0.684 |
| XGBoost        | 3.463 $\pm$ 0.914 | 2.411 $\pm$ 0.163  | 2.466 $\pm$ 0.289 | 2.412 $\pm$ 0.300 | 1.887 $\pm$ 0.624 |
| LightGBM       | 3.421 $\pm$ 0.860 | 2.508 $\pm$ 0.193  | 2.498 $\pm$ 0.414 | 2.558 $\pm$ 0.289 | 1.731 $\pm$ 0.477 |
| DeepCDR (Expr) | 3.803 $\pm$ 1.008 | 2.525 $\pm$ 0.156  | 2.192 $\pm$ 0.310 | 2.137 $\pm$ 0.143 | 2.054 $\pm$ 0.518 |
| DeepTTA        | 3.129 $\pm$ 0.937 | 2.464 $\pm$ 0.151  | 2.233 $\pm$ 0.418 | 2.474 $\pm$ 0.256 | 2.085 $\pm$ 0.528 |
| InFo-DRP       | 2.813 $\pm$ 1.176 | 2.247 $\pm$ 0.150  | 1.974 $\pm$ 0.360 | 1.799 $\pm$ 0.178 | 1.794 $\pm$ 0.435 |

**Table S11. RMSE performance on five non-cancerous cell lines (GDSC<sub>v2</sub>, scaffold-blind setting).**

| Method         | BPH-1          | Hs-940-T       | IM-9           | PWR-1E         | ARH-77         |
|----------------|----------------|----------------|----------------|----------------|----------------|
| Random Forest  | 3.166 (0.4017) | 2.927 (0.4396) | 3.111 (0.4922) | 3.087 (0.4812) | 2.951 (0.5210) |
| Elastic Net    | 2.692 (0.4131) | 2.455 (0.3778) | 2.767 (0.3827) | 2.457 (0.5317) | 2.579 (0.4201) |
| XGBoost        | 2.728 (0.4116) | 2.612 (0.4820) | 2.851 (0.4188) | 2.600 (0.4736) | 2.685 (0.4516) |
| LightGBM       | 2.748 (0.3329) | 2.591 (0.4019) | 2.834 (0.4166) | 2.631 (0.3572) | 2.657 (0.4477) |
| DeepCDR (Expr) | 2.841 (0.5548) | 2.583 (0.4458) | 2.911 (0.6343) | 2.572 (0.6923) | 2.785 (0.6810) |
| DeepTTA        | 2.831 (0.4302) | 2.527 (0.4842) | 2.759 (0.3892) | 2.447 (0.4635) | 2.586 (0.3364) |
| InFo-DRP       | 2.467 (0.4280) | 2.323 (0.4013) | 2.504 (0.4274) | 2.155 (0.5551) | 2.297 (0.4285) |

**Table S12. PCC performance on five non-cancerous cell lines (GDSC<sub>v2</sub>, drug-blind setting).**

| Method         | BPH-1          | Hs-940-T       | IM-9           | PWR-1E         | ARH-77         |
|----------------|----------------|----------------|----------------|----------------|----------------|
| Random Forest  | 0.350 (0.2997) | 0.210 (0.2335) | 0.359 (0.2978) | 0.227 (0.3462) | 0.368 (0.2710) |
| Elastic Net    | 0.415 (0.1966) | 0.330 (0.2315) | 0.492 (0.1750) | 0.388 (0.3034) | 0.489 (0.1641) |
| XGBoost        | 0.439 (0.1877) | 0.307 (0.0981) | 0.498 (0.1283) | 0.395 (0.2189) | 0.503 (0.1558) |
| LightGBM       | 0.419 (0.2569) | 0.241 (0.2099) | 0.457 (0.1878) | 0.328 (0.3082) | 0.467 (0.2367) |
| DeepCDR (Expr) | 0.461 (0.1912) | 0.393 (0.2301) | 0.475 (0.1670) | 0.464 (0.2012) | 0.513 (0.1766) |
| DeepTTA        | 0.394 (0.1372) | 0.272 (0.3126) | 0.460 (0.1313) | 0.414 (0.1583) | 0.436 (0.1693) |
| InFo-DRP       | 0.599 (0.1140) | 0.493 (0.2535) | 0.615 (0.1497) | 0.614 (0.1737) | 0.664 (0.1500) |

**Table S13. RMSE performance on five non-cancerous cell lines (GDSC<sub>v2</sub>, drug-blind setting).**

| Method         | BPH-1          | Hs-940-T       | IM-9           | PWR-1E         | ARH-77         |
|----------------|----------------|----------------|----------------|----------------|----------------|
| Random Forest  | 3.053 (0.7164) | 3.094 (0.8622) | 3.067 (0.8692) | 3.026 (0.6259) | 2.994 (0.7564) |
| Elastic Net    | 2.673 (0.3582) | 2.510 (0.4906) | 2.667 (0.3488) | 2.434 (0.2956) | 2.525 (0.2471) |
| XGBoost        | 2.623 (0.3914) | 2.619 (0.4195) | 2.660 (0.4060) | 2.475 (0.2788) | 2.553 (0.3490) |
| LightGBM       | 2.684 (0.5026) | 2.810 (0.6465) | 2.732 (0.4912) | 2.623 (0.3789) | 2.610 (0.4838) |
| DeepCDR (Expr) | 2.707 (0.4195) | 2.483 (0.5978) | 2.791 (0.3902) | 2.426 (0.4248) | 2.603 (0.4087) |
| DeepTTA        | 2.804 (0.3732) | 2.586 (0.6973) | 2.786 (0.4483) | 2.468 (0.4895) | 2.628 (0.4476) |
| InFo-DRP       | 2.430 (0.3800) | 2.251 (0.5668) | 2.509 (0.5216) | 2.164 (0.5836) | 2.223 (0.4730) |

**Table S14. External validation on CCLE.** All models were trained on GDSC<sub>v2</sub> and evaluated on CCLE without any adaptation. Values are reported as mean (standard deviation) over five training seeds with a fixed data split. Elastic Net is deterministic and therefore shows no variation across seeds. **Bold** values denote the best performance.

| Method         | PCC $\uparrow$       |
|----------------|----------------------|
| Random Forest  | 0.292 (0.005)        |
| Elastic Net    | 0.297 (0.000)        |
| XGBoost        | 0.269 (0.034)        |
| LightGBM       | 0.348 (0.028)        |
| DeepCDR (Expr) | 0.417 (0.021)        |
| DeepTTA        | 0.251 (0.024)        |
| InFo-DRP       | <b>0.431</b> (0.012) |

**Table S15. Statistical comparison of InFo-DRP with baseline models (GDSC<sub>v2</sub>).** Fold-wise results ( $n = 10$ ) were compared using the Wilcoxon signed-rank test. Improvement was defined such that positive values favor InFo-DRP:  $\Delta = \text{PCC}_{\text{InFo-DRP}} - \text{PCC}_{\text{baseline}}$  for PCC and  $\Delta = \text{RMSE}_{\text{baseline}} - \text{RMSE}_{\text{InFo-DRP}}$  for RMSE. Values in the InFo-DRP, Baseline, and Improvement columns are fold means. Benjamini–Hochberg adjusted  $p$ -values are reported.

| Setting        | Metric | Method         | InFo-DRP | Baseline | Improvement ( $\Delta$ ) | adj. $p$ -value       |
|----------------|--------|----------------|----------|----------|--------------------------|-----------------------|
| scaffold-blind | PCC    | Random Forest  | 0.5923   | 0.3131   | +0.2793                  | $4.69 \times 10^{-3}$ |
|                |        | Elastic Net    | 0.5923   | 0.4245   | +0.1679                  | $4.69 \times 10^{-3}$ |
|                |        | XGBoost        | 0.5923   | 0.3980   | +0.1944                  | $4.69 \times 10^{-3}$ |
|                |        | LightGBM       | 0.5923   | 0.4057   | +0.1866                  | $4.69 \times 10^{-3}$ |
|                |        | DeepCDR (Expr) | 0.5923   | 0.4046   | +0.1877                  | $4.69 \times 10^{-3}$ |
|                |        | DeepTTA        | 0.5923   | 0.4355   | +0.1568                  | $3.71 \times 10^{-2}$ |
|                | RMSE   | Random Forest  | 2.2818   | 2.9973   | +0.7155                  | $7.81 \times 10^{-3}$ |
|                |        | Elastic Net    | 2.2818   | 2.5182   | +0.2363                  | $2.34 \times 10^{-2}$ |
|                |        | XGBoost        | 2.2818   | 2.6141   | +0.3323                  | $1.46 \times 10^{-2}$ |
|                |        | LightGBM       | 2.2818   | 2.6411   | +0.3593                  | $7.81 \times 10^{-3}$ |
|                |        | DeepCDR (Expr) | 2.2818   | 2.6404   | +0.3585                  | $7.81 \times 10^{-3}$ |
|                |        | DeepTTA        | 2.2818   | 2.4977   | +0.2158                  | $1.05 \times 10^{-1}$ |
| drug-blind     | PCC    | Random Forest  | 0.6174   | 0.3391   | +0.2783                  | $7.81 \times 10^{-3}$ |
|                |        | Elastic Net    | 0.6174   | 0.4615   | +0.1559                  | $5.86 \times 10^{-3}$ |
|                |        | XGBoost        | 0.6174   | 0.4710   | +0.1465                  | $1.17 \times 10^{-2}$ |
|                |        | LightGBM       | 0.6174   | 0.4366   | +0.1809                  | $2.73 \times 10^{-2}$ |
|                |        | DeepCDR (Expr) | 0.6174   | 0.4886   | +0.1288                  | $8.79 \times 10^{-3}$ |
|                |        | DeepTTA        | 0.6174   | 0.4556   | +0.1619                  | $5.86 \times 10^{-3}$ |
|                | RMSE   | Random Forest  | 2.2405   | 2.9812   | +0.7408                  | $1.76 \times 10^{-2}$ |
|                |        | Elastic Net    | 2.2405   | 2.4856   | +0.2452                  | $5.86 \times 10^{-2}$ |
|                |        | XGBoost        | 2.2405   | 2.5141   | +0.2736                  | $6.45 \times 10^{-2}$ |
|                |        | LightGBM       | 2.2405   | 2.6188   | +0.3783                  | $5.86 \times 10^{-2}$ |
|                |        | DeepCDR (Expr) | 2.2405   | 2.5205   | +0.2801                  | $5.86 \times 10^{-2}$ |
|                |        | DeepTTA        | 2.2405   | 2.5140   | +0.2735                  | $1.76 \times 10^{-2}$ |
| cell-blind     | PCC    | Random Forest  | 0.8900   | 0.8936   | −0.0036                  | $3.71 \times 10^{-2}$ |
|                |        | Elastic Net    | 0.8900   | 0.8703   | +0.0197                  | $5.86 \times 10^{-3}$ |
|                |        | XGBoost        | 0.8900   | 0.8939   | −0.0039                  | $3.71 \times 10^{-2}$ |
|                |        | LightGBM       | 0.8900   | 0.8940   | −0.0040                  | $3.71 \times 10^{-2}$ |
|                |        | DeepCDR (Expr) | 0.8900   | 0.8701   | +0.0199                  | $5.86 \times 10^{-3}$ |
|                |        | DeepTTA        | 0.8900   | 0.8871   | +0.0029                  | $3.71 \times 10^{-2}$ |
|                | RMSE   | Random Forest  | 1.2755   | 1.2479   | −0.0275                  | $1.46 \times 10^{-2}$ |
|                |        | Elastic Net    | 1.2755   | 1.3699   | +0.0944                  | $5.86 \times 10^{-3}$ |
|                |        | XGBoost        | 1.2755   | 1.2461   | −0.0294                  | $1.46 \times 10^{-2}$ |
|                |        | LightGBM       | 1.2755   | 1.2452   | −0.0302                  | $1.64 \times 10^{-2}$ |
|                |        | DeepCDR (Expr) | 1.2755   | 1.4079   | +0.1325                  | $5.86 \times 10^{-3}$ |
|                |        | DeepTTA        | 1.2755   | 1.2916   | +0.0161                  | $1.95 \times 10^{-2}$ |

**Table S16. Statistical comparison of PCC on five non-cancerous cell lines (GDSC<sub>v2</sub>, scaffold-blind).** Fold-wise PCC ( $n = 10$ ) was compared using the Wilcoxon signed-rank test for each cell line. Improvement is defined as  $\Delta = \text{PCC}_{\text{InFo-DRP}} - \text{PCC}_{\text{baseline}}$ , and Benjamini–Hochberg adjusted  $p$ -values are reported.

| Cell line | Method         | InFo-DRP | Baseline | Improvement ( $\Delta$ ) | adj. $p$ -value       |
|-----------|----------------|----------|----------|--------------------------|-----------------------|
| BPH-1     | Random Forest  | 0.5366   | 0.2869   | +0.2496                  | $1.17 \times 10^{-2}$ |
|           | Elastic Net    | 0.5366   | 0.3953   | +0.1413                  | $8.40 \times 10^{-2}$ |
|           | XGBoost        | 0.5366   | 0.3728   | +0.1638                  | $4.10 \times 10^{-2}$ |
|           | LightGBM       | 0.5366   | 0.3709   | +0.1657                  | $4.10 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.5366   | 0.3673   | +0.1693                  | $4.10 \times 10^{-2}$ |
|           | DeepTTA        | 0.5366   | 0.3558   | +0.1808                  | $8.40 \times 10^{-2}$ |
| Hs-940-T  | Random Forest  | 0.4297   | 0.2024   | +0.2272                  | $7.42 \times 10^{-2}$ |
|           | Elastic Net    | 0.4297   | 0.2679   | +0.1617                  | $1.05 \times 10^{-1}$ |
|           | XGBoost        | 0.4297   | 0.1719   | +0.2577                  | $7.42 \times 10^{-2}$ |
|           | LightGBM       | 0.4297   | 0.2566   | +0.1730                  | $7.73 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.4297   | 0.2757   | +0.1540                  | $7.42 \times 10^{-2}$ |
|           | DeepTTA        | 0.4297   | 0.2168   | +0.2129                  | $7.73 \times 10^{-2}$ |
| IM-9      | Random Forest  | 0.5903   | 0.3874   | +0.2029                  | $3.28 \times 10^{-2}$ |
|           | Elastic Net    | 0.5903   | 0.3784   | +0.2119                  | $1.95 \times 10^{-2}$ |
|           | XGBoost        | 0.5903   | 0.3761   | +0.2142                  | $2.05 \times 10^{-2}$ |
|           | LightGBM       | 0.5903   | 0.4175   | +0.1728                  | $1.76 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.5903   | 0.3533   | +0.2370                  | $1.76 \times 10^{-2}$ |
|           | DeepTTA        | 0.5903   | 0.3268   | +0.2635                  | $3.71 \times 10^{-2}$ |
| PWR-1E    | Random Forest  | 0.6050   | 0.1935   | +0.4115                  | $5.86 \times 10^{-3}$ |
|           | Elastic Net    | 0.6050   | 0.3515   | +0.2535                  | $8.79 \times 10^{-3}$ |
|           | XGBoost        | 0.6050   | 0.2826   | +0.3224                  | $1.17 \times 10^{-2}$ |
|           | LightGBM       | 0.6050   | 0.2877   | +0.3173                  | $8.79 \times 10^{-3}$ |
|           | DeepCDR (Expr) | 0.6050   | 0.3468   | +0.2582                  | $5.86 \times 10^{-3}$ |
|           | DeepTTA        | 0.6050   | 0.3349   | +0.2700                  | $1.95 \times 10^{-2}$ |
| ARH-77    | Random Forest  | 0.6125   | 0.4054   | +0.2070                  | $1.93 \times 10^{-1}$ |
|           | Elastic Net    | 0.6125   | 0.3824   | +0.2300                  | $2.93 \times 10^{-2}$ |
|           | XGBoost        | 0.6125   | 0.3696   | +0.2429                  | $2.93 \times 10^{-2}$ |
|           | LightGBM       | 0.6125   | 0.4325   | +0.1799                  | $7.73 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.6125   | 0.3690   | +0.2434                  | $2.34 \times 10^{-2}$ |
|           | DeepTTA        | 0.6125   | 0.3691   | +0.2433                  | $2.93 \times 10^{-2}$ |

**Table S17. Statistical comparison of RMSE on five non-cancerous cell lines (GDSC<sub>v2</sub>, scaffold-blind).** Fold-wise RMSE ( $n = 10$ ) was compared using the Wilcoxon signed-rank test for each cell line. Improvement is defined as  $\Delta = \text{RMSE}_{\text{baseline}} - \text{RMSE}_{\text{InFo-DRP}}$ , and Benjamini–Hochberg adjusted  $p$ -values are reported.

| Cell line | Method         | InFo-DRP | Baseline | Improvement ( $\Delta$ ) | adj. $p$ -value       |
|-----------|----------------|----------|----------|--------------------------|-----------------------|
| BPH-1     | Random Forest  | 2.4666   | 3.1658   | +0.6991                  | $2.34 \times 10^{-2}$ |
|           | Elastic Net    | 2.4666   | 2.6921   | +0.2254                  | $1.31 \times 10^{-1}$ |
|           | XGBoost        | 2.4666   | 2.7284   | +0.2617                  | $7.32 \times 10^{-2}$ |
|           | LightGBM       | 2.4666   | 2.7482   | +0.2815                  | $7.73 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.4666   | 2.8407   | +0.3741                  | $3.91 \times 10^{-2}$ |
|           | DeepTTA        | 2.4666   | 2.8310   | +0.3643                  | $3.91 \times 10^{-2}$ |
| Hs-940-T  | Random Forest  | 2.3231   | 2.9269   | +0.6039                  | $1.17 \times 10^{-2}$ |
|           | Elastic Net    | 2.3231   | 2.4548   | +0.1318                  | $4.32 \times 10^{-1}$ |
|           | XGBoost        | 2.3231   | 2.6122   | +0.2892                  | $7.32 \times 10^{-2}$ |
|           | LightGBM       | 2.3231   | 2.5908   | +0.2678                  | $5.47 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.3231   | 2.5830   | +0.2600                  | $1.76 \times 10^{-2}$ |
|           | DeepTTA        | 2.3231   | 2.5275   | +0.2044                  | $2.32 \times 10^{-1}$ |
| IM-9      | Random Forest  | 2.5037   | 3.1108   | +0.6071                  | $2.73 \times 10^{-2}$ |
|           | Elastic Net    | 2.5037   | 2.7669   | +0.2632                  | $5.86 \times 10^{-2}$ |
|           | XGBoost        | 2.5037   | 2.8512   | +0.3475                  | $2.73 \times 10^{-2}$ |
|           | LightGBM       | 2.5037   | 2.8338   | +0.3301                  | $2.73 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.5037   | 2.9114   | +0.4077                  | $2.93 \times 10^{-2}$ |
|           | DeepTTA        | 2.5037   | 2.7591   | +0.2554                  | $1.05 \times 10^{-1}$ |
| PWR-1E    | Random Forest  | 2.1546   | 3.0873   | +0.9326                  | $1.17 \times 10^{-2}$ |
|           | Elastic Net    | 2.1546   | 2.4565   | +0.3019                  | $1.64 \times 10^{-2}$ |
|           | XGBoost        | 2.1546   | 2.6002   | +0.4456                  | $1.46 \times 10^{-2}$ |
|           | LightGBM       | 2.1546   | 2.6312   | +0.4766                  | $1.17 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.1546   | 2.5719   | +0.4172                  | $1.17 \times 10^{-2}$ |
|           | DeepTTA        | 2.1546   | 2.4474   | +0.2928                  | $6.45 \times 10^{-2}$ |
| ARH-77    | Random Forest  | 2.2970   | 2.9507   | +0.6536                  | $3.91 \times 10^{-2}$ |
|           | Elastic Net    | 2.2970   | 2.5788   | +0.2817                  | $3.91 \times 10^{-2}$ |
|           | XGBoost        | 2.2970   | 2.6853   | +0.3883                  | $8.40 \times 10^{-2}$ |
|           | LightGBM       | 2.2970   | 2.6572   | +0.3601                  | $5.57 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.2970   | 2.7854   | +0.4883                  | $3.91 \times 10^{-2}$ |
|           | DeepTTA        | 2.2970   | 2.5863   | +0.2893                  | $5.86 \times 10^{-2}$ |

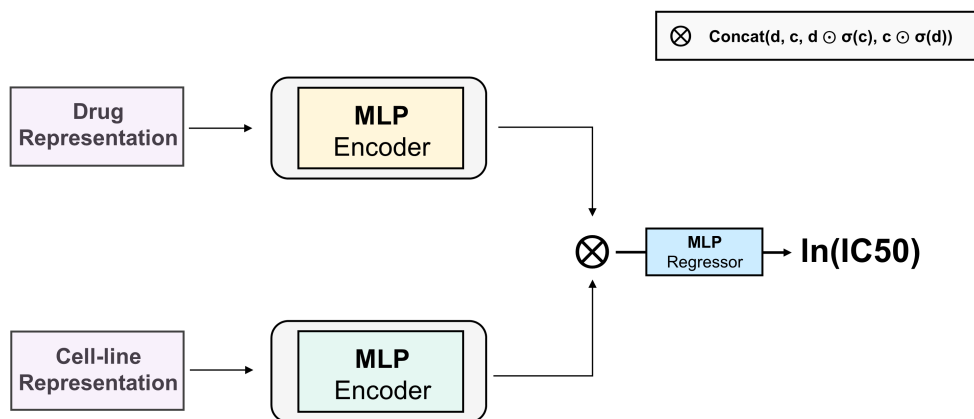
**Table S18. Statistical comparison of PCC on five non-cancerous cell lines (GDSC<sub>v2</sub>, drug-blind).** Fold-wise PCC ( $n = 10$ ) was compared using the Wilcoxon signed-rank test for each cell line. Improvement is defined as  $\Delta = \text{PCC}_{\text{InFo-DRP}} - \text{PCC}_{\text{baseline}}$ , and Benjamini–Hochberg adjusted  $p$ -values are reported.

| Cell line | Method         | InFo-DRP | Baseline | Improvement ( $\Delta$ ) | adj. $p$ -value       |
|-----------|----------------|----------|----------|--------------------------|-----------------------|
| BPH-1     | Random Forest  | 0.5991   | 0.3500   | +0.2491                  | $2.73 \times 10^{-2}$ |
|           | Elastic Net    | 0.5991   | 0.4153   | +0.1838                  | $2.73 \times 10^{-2}$ |
|           | XGBoost        | 0.5991   | 0.4387   | +0.1604                  | $2.93 \times 10^{-2}$ |
|           | LightGBM       | 0.5991   | 0.4185   | +0.1806                  | $6.45 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.5991   | 0.4610   | +0.1381                  | $6.45 \times 10^{-2}$ |
|           | DeepTTA        | 0.5991   | 0.3935   | +0.2056                  | $1.17 \times 10^{-2}$ |
| Hs-940-T  | Random Forest  | 0.4928   | 0.2097   | +0.2831                  | $9.67 \times 10^{-2}$ |
|           | Elastic Net    | 0.4928   | 0.3300   | +0.1628                  | $1.01 \times 10^{-1}$ |
|           | XGBoost        | 0.4928   | 0.3072   | +0.1856                  | $9.67 \times 10^{-2}$ |
|           | LightGBM       | 0.4928   | 0.2408   | +0.2519                  | $9.67 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.4928   | 0.3925   | +0.1003                  | $1.60 \times 10^{-1}$ |
|           | DeepTTA        | 0.4928   | 0.2723   | +0.2205                  | $9.67 \times 10^{-2}$ |
| IM-9      | Random Forest  | 0.6149   | 0.3587   | +0.2562                  | $4.10 \times 10^{-2}$ |
|           | Elastic Net    | 0.6149   | 0.4925   | +0.1224                  | $3.91 \times 10^{-2}$ |
|           | XGBoost        | 0.6149   | 0.4978   | +0.1171                  | $4.88 \times 10^{-2}$ |
|           | LightGBM       | 0.6149   | 0.4575   | +0.1574                  | $3.91 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.6149   | 0.4754   | +0.1395                  | $3.91 \times 10^{-2}$ |
|           | DeepTTA        | 0.6149   | 0.4604   | +0.1545                  | $4.45 \times 10^{-2}$ |
| PWR-1E    | Random Forest  | 0.6139   | 0.2269   | +0.3870                  | $2.93 \times 10^{-2}$ |
|           | Elastic Net    | 0.6139   | 0.3881   | +0.2258                  | $4.88 \times 10^{-2}$ |
|           | XGBoost        | 0.6139   | 0.3947   | +0.2192                  | $4.88 \times 10^{-2}$ |
|           | LightGBM       | 0.6139   | 0.3281   | +0.2858                  | $4.88 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.6139   | 0.4644   | +0.1495                  | $4.88 \times 10^{-2}$ |
|           | DeepTTA        | 0.6139   | 0.4136   | +0.2003                  | $2.34 \times 10^{-2}$ |
| ARH-77    | Random Forest  | 0.6642   | 0.3676   | +0.2966                  | $2.05 \times 10^{-2}$ |
|           | Elastic Net    | 0.6642   | 0.4888   | +0.1754                  | $1.95 \times 10^{-2}$ |
|           | XGBoost        | 0.6642   | 0.5027   | +0.1615                  | $3.28 \times 10^{-2}$ |
|           | LightGBM       | 0.6642   | 0.4671   | +0.1971                  | $4.88 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 0.6642   | 0.5132   | +0.1510                  | $1.76 \times 10^{-2}$ |
|           | DeepTTA        | 0.6642   | 0.4355   | +0.2287                  | $1.17 \times 10^{-2}$ |

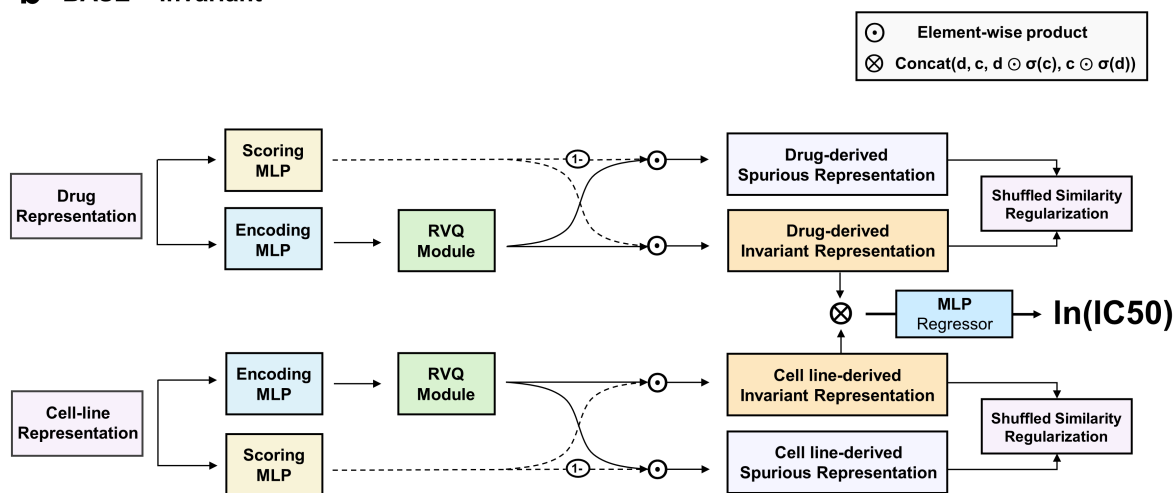
**Table S19. Statistical comparison of RMSE on five non-cancerous cell lines (GDSC<sub>v2</sub>, drug-blind).** Fold-wise RMSE ( $n = 10$ ) was compared using the Wilcoxon signed-rank test for each cell line. Improvement is defined as  $\Delta = \text{RMSE}_{\text{baseline}} - \text{RMSE}_{\text{InFo-DRP}}$ , and Benjamini–Hochberg adjusted  $p$ -values are reported.

| Cell line | Method         | InFo-DRP | Baseline | Improvement ( $\Delta$ ) | adj. $p$ -value       |
|-----------|----------------|----------|----------|--------------------------|-----------------------|
| BPH-1     | Random Forest  | 2.4298   | 3.0531   | +0.6233                  | $4.10 \times 10^{-2}$ |
|           | Elastic Net    | 2.4298   | 2.6734   | +0.2436                  | $1.26 \times 10^{-1}$ |
|           | XGBoost        | 2.4298   | 2.6230   | +0.1933                  | $2.32 \times 10^{-1}$ |
|           | LightGBM       | 2.4298   | 2.6837   | +0.2539                  | $2.75 \times 10^{-1}$ |
|           | DeepCDR (Expr) | 2.4298   | 2.7073   | +0.2775                  | $1.26 \times 10^{-1}$ |
|           | DeepTTA        | 2.4298   | 2.8040   | +0.3742                  | $4.10 \times 10^{-2}$ |
| Hs-940-T  | Random Forest  | 2.2505   | 3.0937   | +0.8432                  | $1.17 \times 10^{-2}$ |
|           | Elastic Net    | 2.2505   | 2.5103   | +0.2598                  | $1.05 \times 10^{-1}$ |
|           | XGBoost        | 2.2505   | 2.6190   | +0.3685                  | $4.10 \times 10^{-2}$ |
|           | LightGBM       | 2.2505   | 2.8101   | +0.5596                  | $4.10 \times 10^{-2}$ |
|           | DeepCDR (Expr) | 2.2505   | 2.4826   | +0.2321                  | $5.86 \times 10^{-2}$ |
|           | DeepTTA        | 2.2505   | 2.5861   | +0.3356                  | $4.10 \times 10^{-2}$ |
| IM-9      | Random Forest  | 2.5094   | 3.0669   | +0.5575                  | $1.68 \times 10^{-1}$ |
|           | Elastic Net    | 2.5094   | 2.6666   | +0.1572                  | $2.32 \times 10^{-1}$ |
|           | XGBoost        | 2.5094   | 2.6596   | +0.1502                  | $3.75 \times 10^{-1}$ |
|           | LightGBM       | 2.5094   | 2.7316   | +0.2222                  | $2.32 \times 10^{-1}$ |
|           | DeepCDR (Expr) | 2.5094   | 2.7910   | +0.2816                  | $1.68 \times 10^{-1}$ |
|           | DeepTTA        | 2.5094   | 2.7857   | +0.2763                  | $1.68 \times 10^{-1}$ |
| PWR-1E    | Random Forest  | 2.1644   | 3.0264   | +0.8620                  | $1.17 \times 10^{-1}$ |
|           | Elastic Net    | 2.1644   | 2.4343   | +0.2698                  | $1.60 \times 10^{-1}$ |
|           | XGBoost        | 2.1644   | 2.4749   | +0.3105                  | $1.27 \times 10^{-1}$ |
|           | LightGBM       | 2.1644   | 2.6232   | +0.4588                  | $1.26 \times 10^{-1}$ |
|           | DeepCDR (Expr) | 2.1644   | 2.4256   | +0.2611                  | $1.26 \times 10^{-1}$ |
|           | DeepTTA        | 2.1644   | 2.4675   | +0.3031                  | $1.26 \times 10^{-1}$ |
| ARH-77    | Random Forest  | 2.2226   | 2.9937   | +0.7711                  | $2.73 \times 10^{-2}$ |
|           | Elastic Net    | 2.2226   | 2.5250   | +0.3024                  | $7.32 \times 10^{-2}$ |
|           | XGBoost        | 2.2226   | 2.5530   | +0.3304                  | $7.73 \times 10^{-2}$ |
|           | LightGBM       | 2.2226   | 2.6098   | +0.3872                  | $1.05 \times 10^{-1}$ |
|           | DeepCDR (Expr) | 2.2226   | 2.6035   | +0.3809                  | $1.76 \times 10^{-2}$ |
|           | DeepTTA        | 2.2226   | 2.6284   | +0.4058                  | $1.76 \times 10^{-2}$ |

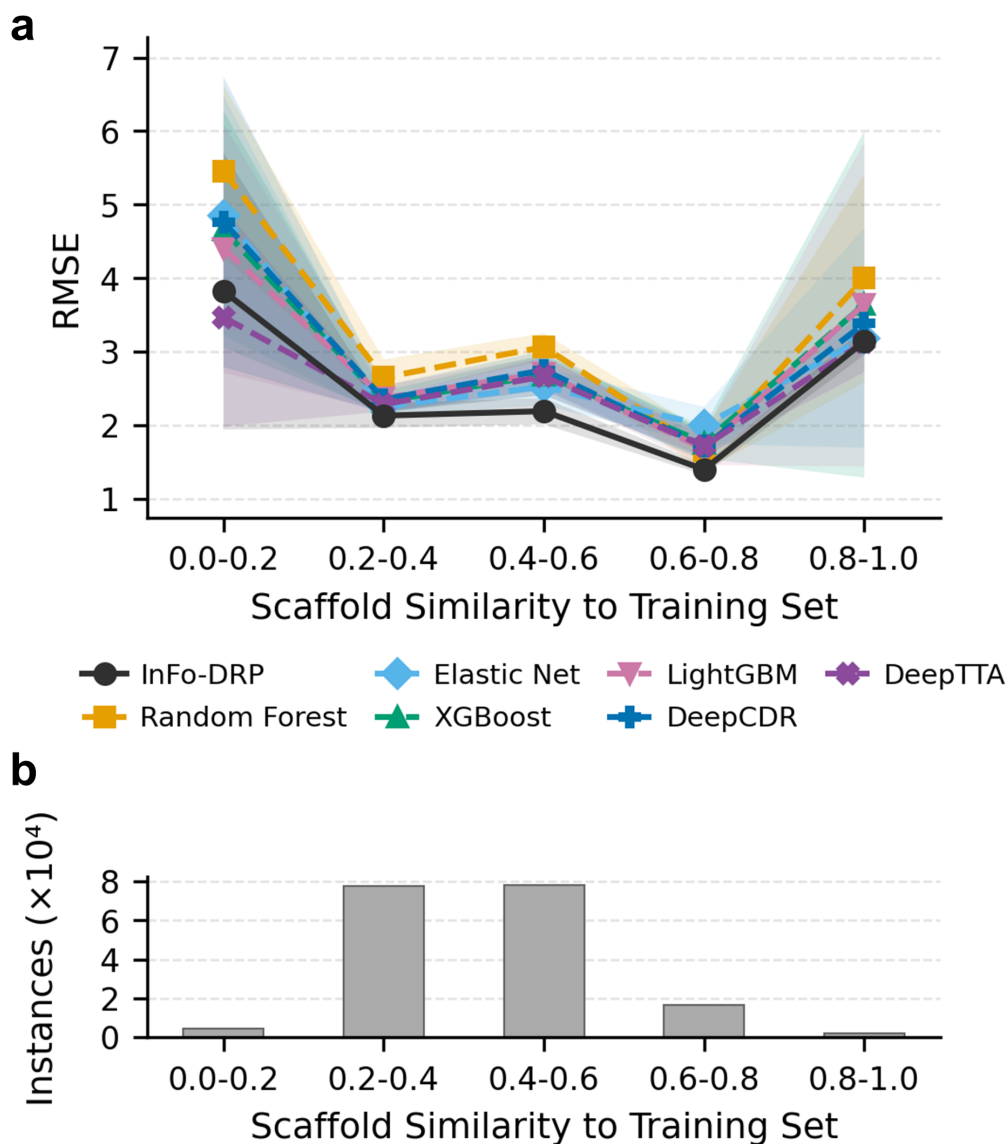
**a BASE (no Invariant)**



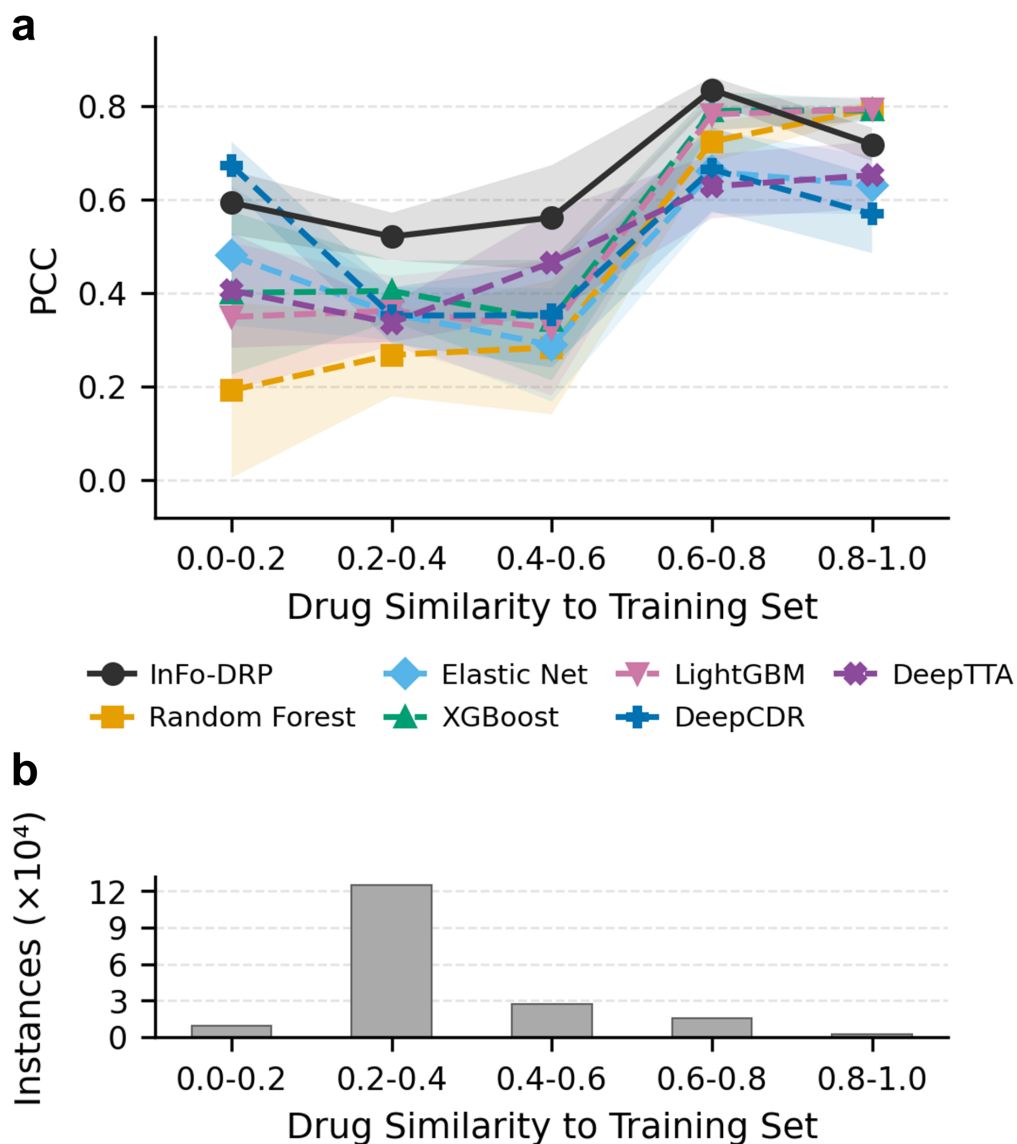
**b BASE + Invariant**



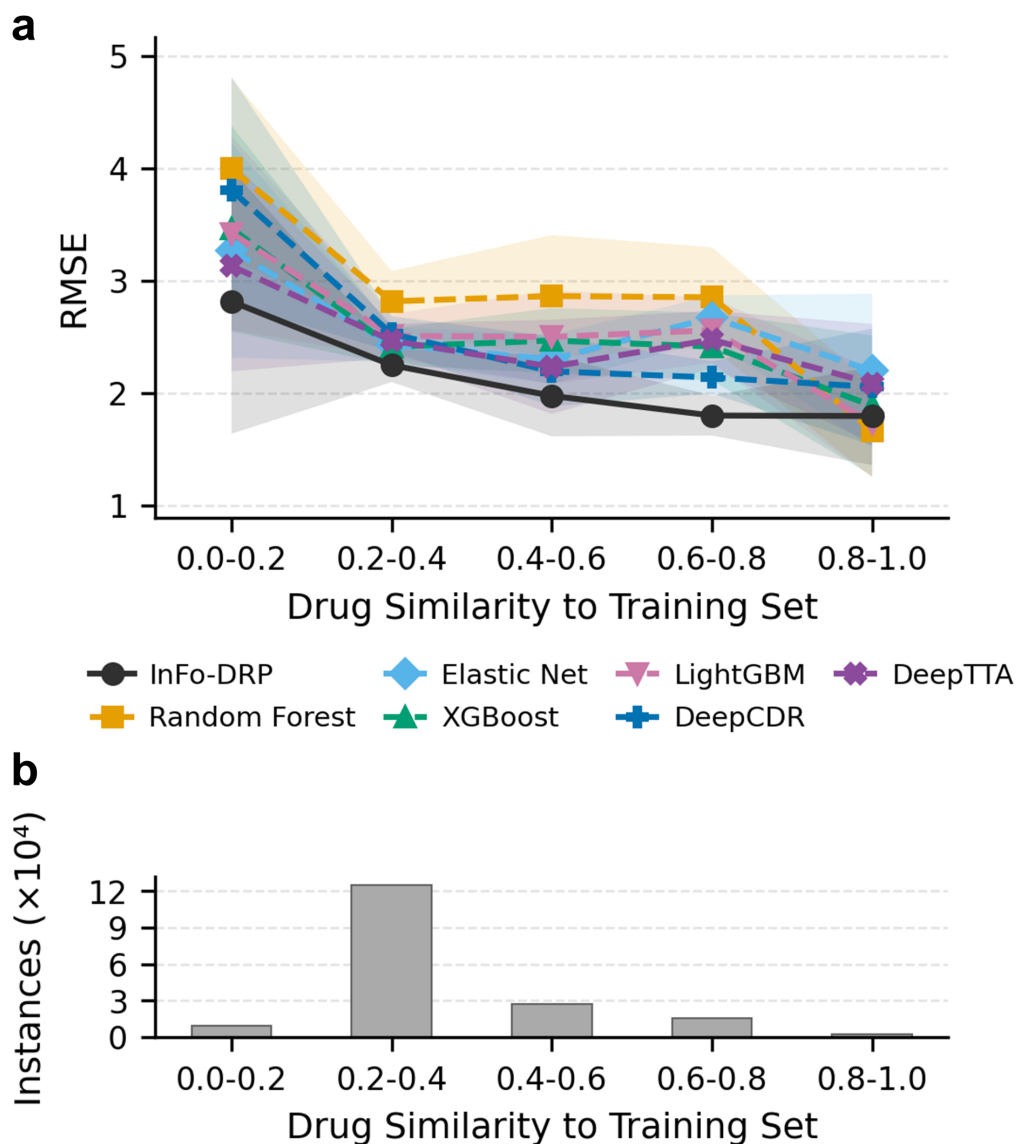
**Figure S1. Architectures of single-branch BASE variants used in the ablation analysis.** (a) BASE architecture without invariant feature extraction, where a single drug representation (MolBERT, Chemformer, ECFP4, or physicochemical descriptors) is fused with the cell-line gene expression representation to predict  $\ln(\text{IC}_{50})$ . (b) BASE architecture with invariant feature extraction, in which drug and cell representations are decomposed into invariant and spurious components by element-wise gates (dashed lines). Only the invariant representations are used for downstream prediction, while spurious components are excluded from the prediction pathway.



**Figure S2. Scaffold-level generalization (RMSE; GDSC<sub>v2</sub>, scaffold-blind setting).** (a) Mean RMSE across scaffold similarity bins, defined by the maximum ECFP4 Tanimoto similarity between Bemis–Murcko scaffolds in the test set and those in the training set; shaded regions indicate the standard error of the mean (SEM) across folds. (b) Number of test instances per similarity bin.



**Figure S3. Drug-level generalization (PCC; GDSC<sub>v2</sub>, drug-blind setting).** (a) Mean PCC across drug similarity bins, defined by the maximum ECFP4 Tanimoto similarity between test drugs and those in the training set; shaded regions indicate the standard error of the mean (SEM) across folds. (b) Number of test instances per similarity bin.



**Figure S4. Drug-level generalization (RMSE; GDSC<sub>v2</sub>, drug-blind setting).** (a) Mean RMSE across drug similarity bins, defined by the maximum ECFP4 Tanimoto similarity between test drugs and those in the training set; shaded regions indicate the standard error of the mean (SEM) across folds. (b) Number of test instances per similarity bin.